BURJ KHALIFA

An Encyclopedia of Its Construction

Origins • Purpose • Timeline • Key Events • Completion • Key Figures

Dubai, United Arab Emirates
Construction: January 2004 – January 2010
Height: 828 metres (2,717 feet)

Compiled October 2026 from published reporting, the tower's official website, and reference sources.

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1. At a Glance

Item	Detail
Former name	Burj Dubai (“Dubai Tower”), used until the official opening
Location	Downtown Dubai, along Sheikh Zayed Road, Dubai, UAE
Developer / owner	Emaar Properties
Architect	Adrian Smith, Skidmore, Owings & Merrill (SOM), Chicago
Structural engineer	Bill Baker, SOM
Main contractor	Joint venture of Samsung C&T (South Korea), BESIX (Belgium) and Arabtec (UAE)
Project manager	Turner Construction International
Construction began	January 2004 (excavation); piling from February 2004
Exterior completed	1 October 2009
Official opening	4 January 2010
Fully operational	23 January 2011, when the last major planned facility opened
Height	828 m (2,717 ft); more than 160 floors
Reported cost	Approximately US$1.5 billion
Elevators / escalators	57 elevators and 8 escalators

The Burj Khalifa is a skyscraper of mixed use, with residences, offices and a hotel. When it opened it was the tallest building and the tallest man-made structure in the world, and it overtook Taipei 101 (about 509 m), which had held the building record until then.

2. Why the Tower Was Built

Economic diversification. Dubai has far smaller oil reserves than its neighbour Abu Dhabi, and from the late 1990s its leadership pursued growth through trade, tourism, aviation, finance and real estate. Landmark projects, such as the sail-shaped Burj Al Arab hotel and the Palm Jumeirah island, were part of this strategy. A record-breaking tower was the next step.

Putting Dubai on the map. An executive from Nakheel Properties recalled that Dubai's ruler, Sheikh Mohammed bin Rashid Al Maktoum, wanted to put the emirate on the map with something truly sensational. Officials also argued that projects of this kind were needed to win international recognition and, with it, foreign investment.

The centrepiece of a new district. The tower was designed as the anchor of Downtown Dubai (originally “Downtown Burj Dubai”), a large mixed-use development planned to include around 30,000 homes, parkland, a large artificial lake, and the Dubai Mall, one of the world's largest shopping centres. The tower was meant to lift the value of everything around it.

A planned first. In its early planning the tower was imagined as entirely residential. It later became a mix of residences (including Armani Residences), offices and a hotel, and the finishes were upgraded as the project progressed.

Why it was renamed. The tower was always intended to carry the name “Burj Dubai”. The name changed on opening day for reasons tied to the global financial crisis; see Section 9.

3. Design and Planning

The design was produced by Skidmore, Owings & Merrill, the Chicago firm that had also designed the Sears (now Willis) Tower, a former world-record holder. Adrian Smith led the architecture, and Bill Baker led the structural engineering.

Inspiration. The three-lobed, Y-shaped floor plan is credited to the Hymenocallis, a desert flower, and to patterns in Islamic architecture. Smith has said that he found a prototype for the plan in the curved three-wing layout of Lake Point Tower in Chicago, visible from his office window. Reports also point to Tower Palace Three, a residential building in Seoul, as a structural precedent.

Buttressed core. Wind was the dominant design problem. SOM developed a “buttressed core” system: a six-sided central core that resists twisting, flanked by three wings that resist wind shear and brace one another. The wings step back in a spiral as they rise, which disrupts wind vortices. More than 40 wind-tunnel tests were run (at the RWDI laboratory in Guelph, Canada), and even the construction cranes were tested.

Design changes. The architect reportedly disliked the first design of the upper part of the tower, and Adrian Smith left SOM in 2006 to found his own firm, before the tower topped out.

4. When Construction Started

Excavation began in **January 2004**, with the first bored piles started in **February 2004**. Sources differ on the exact day: some give 6 January 2004, others 12 January 2004, and others cite 21 September 2004 as the date on which the main contractors formally began or a groundbreaking was marked. The best-supported reading is that site work started in January 2004 and that the main vertical construction (the superstructure) began in **March 2005**.

The foundation work was substantial. The tower rests on a reinforced-concrete mat supported by 192 bored piles. The piles are about 1.5 m in diameter, and published lengths range from roughly 43 m to about 50 m. A high-density, low-permeability concrete was specified, and a cathodic protection system was installed under the mat because of corrosive chemicals in Dubai's groundwater.

5. How Long It Took

Measure	Duration
Start of site work to official opening	About 6 years (January 2004 to 4 January 2010)
Start of site work to exterior completion	About 5 years 9 months (January 2004 to 1 October 2009)
Start of superstructure to topping out	About 3 years 10 months (March 2005 to January 2009)
Start of site work to full operation	About 7 years (January 2004 to 23 January 2011)

The climb was fast at its peak: commonly cited figures put the rate at roughly one floor every three days at its fastest. The tower was originally targeted for 2008, but delays moved the exterior completion to 2009 and the opening to 2010.

6. Construction Timeline

Date	Milestone
January 2004	Excavation begins at the site.
February 2004	Piling begins.
21 September 2004	Date cited by several sources for the formal start of main construction by the contractors.
March 2005	Superstructure begins to rise above ground.
21 – 22 March 2006	Labour unrest on the project's outskirts (see Section 7).
June 2006	Level 50 reached.
January 2007	Level 100 reached.
March – May 2007	Levels 110, 120 and 130 reached in turn.
May 2007	Exterior cladding begins.
21 July 2007	Level 141 reached; the tower becomes the world's tallest building, passing Taipei 101.
September 2007	Level 150 reached; becomes the tallest free-standing structure, passing the CN Tower.
27 October 2007	Strike by thousands of workers (see Section 7).
November 2007	Concrete pumped to a record 601 m.
April 2008	Level 160 reached; becomes the tallest man-made structure ever built.
9 June 2008	Emaar announces a delay to September 2009 due to upgraded finishes.
January 2009	Spire completed; the tower tops out at 828 m.
September 2009	Exterior cladding finished.
1 October 2009	Exterior of the structure complete.
4 January 2010	Official opening; renamed Burj Khalifa.
23 January 2011	Last major planned facility opens; tower fully operational.

Note: the 21 July 2007 and September 2007 dates for passing Taipei 101 and the CN Tower follow Emaar's contemporary announcements and standard references.

7. Key Events During Construction

The Record-Breaking Climb

The tower broke three records on its way up. In July 2007 it passed Taipei 101 to become the tallest building. In September 2007 it passed the CN Tower as the tallest free-standing structure. In April 2008 it surpassed a television mast in North Dakota to become the tallest man-made structure ever built. By the time the spire was finished in January 2009, the tower stood roughly 320 m taller than Taipei 101.

Labour Unrest, March 2006

On the night of 21 March 2006, around 2,500 workers employed by the Dubai firm Al Naboodah Laing O'Rourke, on buildings and housing around the Burj Dubai site, rioted over pay, unpaid overtime and treatment by foremen. They damaged cars, offices, computers and machinery, with damage estimated at about US$1 million. The workers stopped work the next day too, and the Emirates' labour authorities intervened. The main tower joint venture said the disturbances took place on the outskirts and not in the tower's own construction area, and that the tower's progress was unaffected. Soon afterwards the government announced it would allow construction unions.

Strike, October 2007

On 27 October 2007, more than 4,500 workers on the Burj Dubai, most of them Indian, downed tools over pay and conditions, and the dispute continued into November. The owners later said the strike was over.

The Concrete-Pumping Record, November 2007

In November 2007, high-strength concrete (80 megapascal) was pumped from ground level to a height of 601 m, breaking the previous record held for a building by Taipei 101 (470 m) and the overall vertical-pumping record of 532 m. Pressure during pumping was nearly 200 bar.

Delay for Upgraded Finishes, June 2008

On 9 June 2008, Emaar announced that completion would slip to September 2009. A spokesperson explained that the luxury finishes chosen in 2004, when the tower was first conceived, were being replaced with upgraded ones.

The Global Financial Crisis, 2008 to 2010

The tower's later stages unfolded as Dubai's property boom collapsed. Many projects were suspended or abandoned, but the Burj kept going. By late 2009 Dubai was in deep financial trouble, with debts reported at around US$100 billion at the state-linked level, and Abu Dhabi provided about US$25 billion of direct and indirect support in 2009, including a US$10 billion rescue in December.

8. Engineering Highlights

Feature	Detail
Structural system	Buttressed core: hexagonal central core with three wings in a Y plan.
Material	Reinforced concrete up to about level 156; structural steel in the spire above.
Foundation	Concrete mat on 192 bored piles, each about 1.5 m wide, roughly 43 to 50 m deep.
Concrete in foundation	Nearly 45,000 cubic metres, weighing over 110,000 tonnes.
Concrete pumping	Two of the world's largest pumps; pumped to 601 m, a world record at the time.
Facade	Roughly 26,000 individually cut glass panels; the highest installation of an aluminium and glass facade in the world, at 512 m.
Wind testing	More than 40 wind-tunnel tests; setbacks tuned to confuse wind vortices.
Spire	Telescopic spire that is part of the structure and houses communications equipment.
Safety	Fireproof concrete-reinforced stairwells and refuge floors.

Dubai's heat made concrete placement a particular challenge: the mixes had to resist separating under the huge pumping pressure and avoid setting too quickly in summer temperatures. Pours were often done at night, and ice was used to cool the mix.

9. Completion, Opening and Renaming

Exterior completion. The exterior of the structure was complete on 1 October 2009, after the cladding had been finished in September 2009.

Opening ceremony, 4 January 2010. The tower was opened in a ceremony staged by Prisme International. A 72,000-watt light projector lit the tower, fountains and fireworks depicted the Hymenocallis flower, and a lighting sequence imitating a beating heart marked the construction phase. Spotlights reached about 10 km into the sky, and the show ended with fireworks. Screens revealed the tower's long-secret height to the crowd for the first time: 828 m.

The renaming. At the ceremony, Dubai's ruler unexpectedly announced that the tower would be called Burj Khalifa, after Sheikh Khalifa bin Zayed Al Nahyan, ruler of Abu Dhabi and President of the UAE. Commentators widely read the gesture as thanks for Abu Dhabi's financial rescue of Dubai during the debt crisis.

Unfinished interiors. Much of the building was still unfinished inside at the opening. The tower became fully operational on 23 January 2011 with the opening of the last major planned facility.

Records. On completion it was the world's tallest building, tallest free-standing structure and tallest man-made structure. It also held records for the highest installation of an aluminium and glass facade, and for the highest vertical concrete pumping.

10. Key Figures and Organisations

Name	Role
Sheikh Mohammed bin Rashid Al Maktoum	Ruler of Dubai; the driving political vision behind the project; announced the renaming.
Sheikh Khalifa bin Zayed Al Nahyan	Ruler of Abu Dhabi and UAE President; the tower is named in his honour after Abu Dhabi's financial support.
Mohamed Alabbar	Chairman of Emaar Properties, the developer; the public face of the project.
Emaar Properties	Developer and owner of the tower and Downtown Dubai.
Adrian Smith	Design architect at SOM; conceived the Y-shaped form.
Bill Baker	Chief structural engineer at SOM; led the buttressed-core system.
Skidmore, Owings & Merrill (SOM)	Architecture and structural engineering.
Samsung C&T	Lead contractor, South Korea; also built Taipei 101 and the Petronas Towers.
BESIX	Belgian construction group in the main contractor joint venture.
Arabtec	UAE contractor in the main joint venture.
Turner Construction International	Project manager.
Hyder Consulting	Supervision of the works on site.
RWDI (Guelph, Canada)	Wind-tunnel testing of the tower and the cranes.
Prisme International	Producer of the opening ceremony.

Some references also name NASA Multiplex as the foundation contractor; this is reported less consistently than the other roles above.

11. The Workforce

Building the tower took more than 22 million man-hours from about 12,000 professionals and skilled workers and more than 60 consultants and contracting companies from around the world, according to the project's published figures. The great majority of manual workers were migrants from South Asia, as in the wider UAE construction industry.

Their conditions drew sustained criticism. A Human Rights Watch report of November 2006 on construction workers in the UAE described unpaid or very low wages, debt to recruitment agencies, the withholding of passports and hazardous working conditions. That report put the average migrant construction worker's earnings at about US$175 a month. The Dubai government rejected accusations of systematic injustice, but announced measures in March 2006 to allow construction unions. Workers' protests in 2006 and 2007, described in Section 7, are part of the tower's recorded history, and any complete account of the project includes them alongside the engineering.

12. Notes on Sources and Conflicting Dates

The facts above are drawn from Emaar's official Burj Khalifa website, Wikipedia's article on the tower and its labour-force article, Associated Press and NBC News reporting on the opening, Reuters and Khaleej Times coverage of the 2006 unrest, a Mail & Guardian analysis of the opening, the Swarthmore Global Nonviolent Action Database, Arabian Business's construction feature, Concrete Products magazine, MEED, and a set of project fact sheets.

Where sources conflict: *(1) The start date is given variously as 6 January, 12 January and 21 September 2004. (2) Some pages state that the opening was when "the final steel beam was installed"; that claim is not supported by better sources, which date topping out to January 2009. (3) Piles are reported between about 43 m and 50 m long depending on whether the tower or the podium is meant. (4) Costs are reported as about US$1.5 billion; the broader Downtown development cost far more. The figures in this document use the best-supported version in each case.*

Please verify any figure you plan to cite in formal work against Emaar's own published material or the Council on Tall Buildings and Urban Habitat (CTBUH).